

Secretary-style split search for decision trees

Mathias Valla

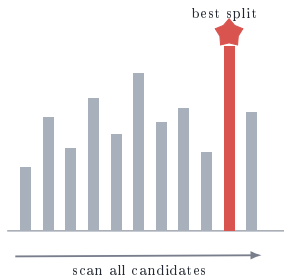
How far can secretary-style early stopping replace exhaustive split search in decision trees?

We compare three within-node strategies over the same candidate space to study the trade-off between training time and predictive performance.

Node

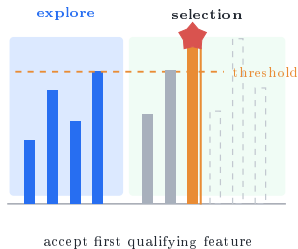
Exhaustive CART

Evaluate every admissible covariate-threshold pair



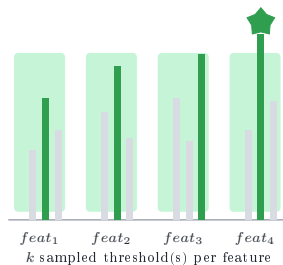
Secretary-style

Explore briefly, set a threshold, accept the first feature whose best split clears it



ERT-style random thresholds

Sample k random threshold(s) per feature then choose the best sampled split



Main idea: test whether online early stopping can approximate exhaustive CART split search.

Trade-off: exhaustive search gives the best split; secretary-style search stops early after exploration; ERT samples very few thresholds and is strongly speed-oriented.